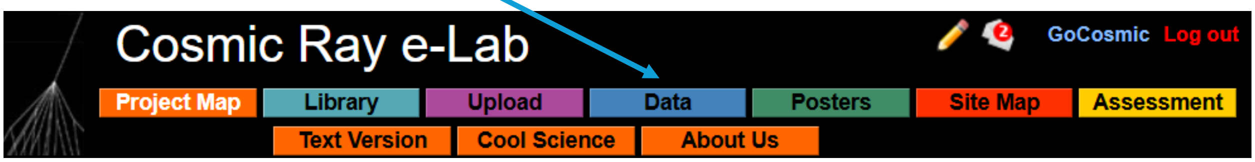
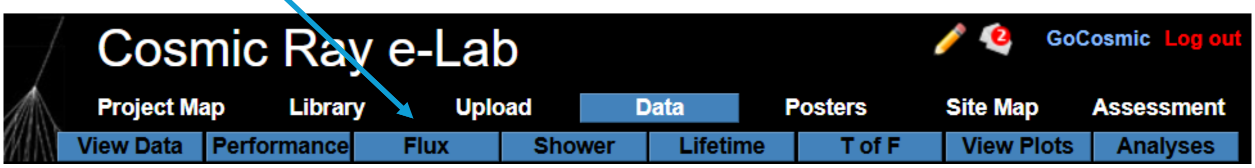
eLab Barometric Pressure Correction

A control month with no significant geomagnetic event (Kp-Index <4) occurs is needed. This is required as a baseline month where the only effect on flux is due to the normal barometric pressure variations. Using the Cosmic Ray eLab *analysis.out* tool to export data into a spreadsheet.

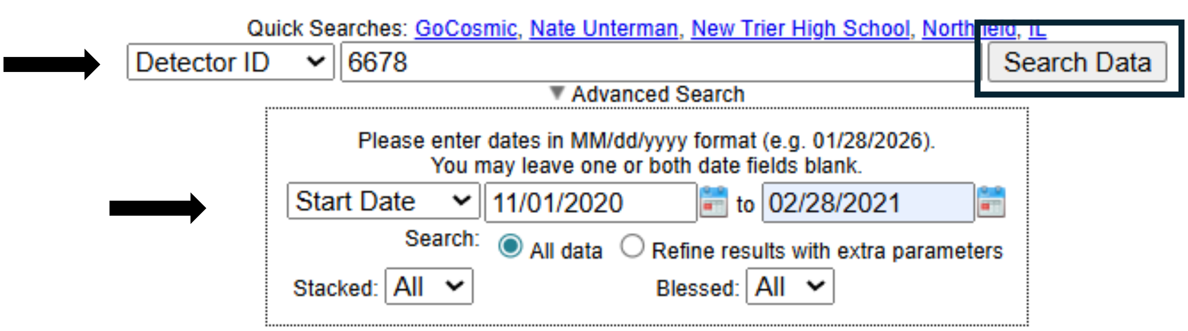
From the landing page, select **Data**.



Select the **Flux** tool.



Select the *Detector ID* from the drop-down menu, Enter the DAQ Number you want for analysis. Using the ► *Advanced Search* to select the month of data you require as a baseline. In this example, we are using *Detector ID* **6678** for *Start Date* **11/01/2020** ***to*** **02/28/2021** (November 2020 through February 2021***). NOTE: Try not to do analysis for more than 6 weeks. The server can become overloaded***. Then press **Search Data**.



Open the school and month desired. We are selecting Sunday 01 November 2020 to Thursday 31 December 2020. If you are starting on a partial day (say, Thursday 05.November), move to the next fill day (here, Friday 06.November). This makes the analysis much easier since we begin on a complete day starting at about midnight UTC, and then go with contiguous files. Note that within this data set, there may be partial days you wish to omit. This is the place to omit files.

New Trier High School
Northfield, IL
150 data files: 0 blessed, 150 stacked, 151,435,340 total events.

November 2020, 38 files Select: All None

Detector 6678, 38 files Select: All None

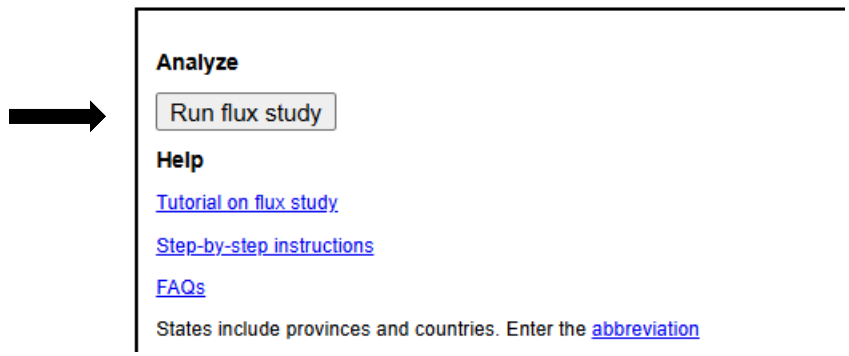
Sun 01 1,300,999 events	Mon 02 1,271,159 events	Tue 03 1,277,592 events	Wed 04 1,277,304 events
Thu 05 855,393 events	Thu 05 422,721 events	Fri 06 1,270,100 events	Sat 07 1,267,810 events
Sun 08 1,266,750 events	Mon 09 1,223,983 events	Mon 09 49,423 events	Tue 10 1,295,649 events
Wed 11 1,293,171 events	Thu 12 703,546 events	Thu 12 577,297 events	Fri 13 1,287,309 events
Sat 14 1,288,170 events	Sun 15 1,332,781 events	Mon 16 733,468 events	Mon 16 562,693 events
Tue 17 1,278,255 events	Wed 18 1,252,417 events	Thu 19 713,988 events	Thu 19 565,922 events
Fri 20 1,271,624 events	Sat 21 1,251,371 events	Sun 22 1,276,971 events	Mon 23 786,486 events
Mon 23 417,442 events	Tue 24 1,276,912 events	Wed 25 1,307,791 events	Thu 26 1,303,881 events
Fri 27 687,988 events	Fri 27 604,998 events	Sat 28 1,287,110 events	Sun 29 1,295,075 events
Mon 30 777,372 events	Mon 30 529,148 events		

December 2020, 36 files Select: All None

Detector 6678, 36 files Select: All None

Tue 01 1,300,401 events	Wed 02 1,282,255 events	Thu 03 736,635 events	Thu 03 549,626 events
Fri 04 1,298,731 events	Sat 05 1,287,499 events	Sun 06 1,287,163 events	Mon 07 764,688 events
Mon 07 527,029 events	Tue 08 1,291,527 events	Wed 09 1,304,354 events	Thu 10 691,006 events
Thu 10 591,585 events	Fri 11 1,295,893 events	Sat 12 1,313,546 events	Sun 13 984,360 events
Sun 13 304,917 events	Mon 14 1,289,039 events	Tue 15 1,271,769 events	Wed 16 1,236,627 events
Wed 16 68,507 events	Thu 17 1,305,168 events	Fri 18 1,282,785 events	Sat 19 1,296,541 events
Sun 20 1,312,944 events	Mon 21 1,332,665 events	Tue 22 202,856 events	Tue 22 1,092,393 events
Wed 23 1,312,917 events	Thu 24 1,324,975 events	Fri 25 812,574 events	Mon 28 596,156 events
Tue 29 726,460 events	Tue 29 528,808 events	Wed 30 1,302,039 events	Thu 31 1,287,982 events

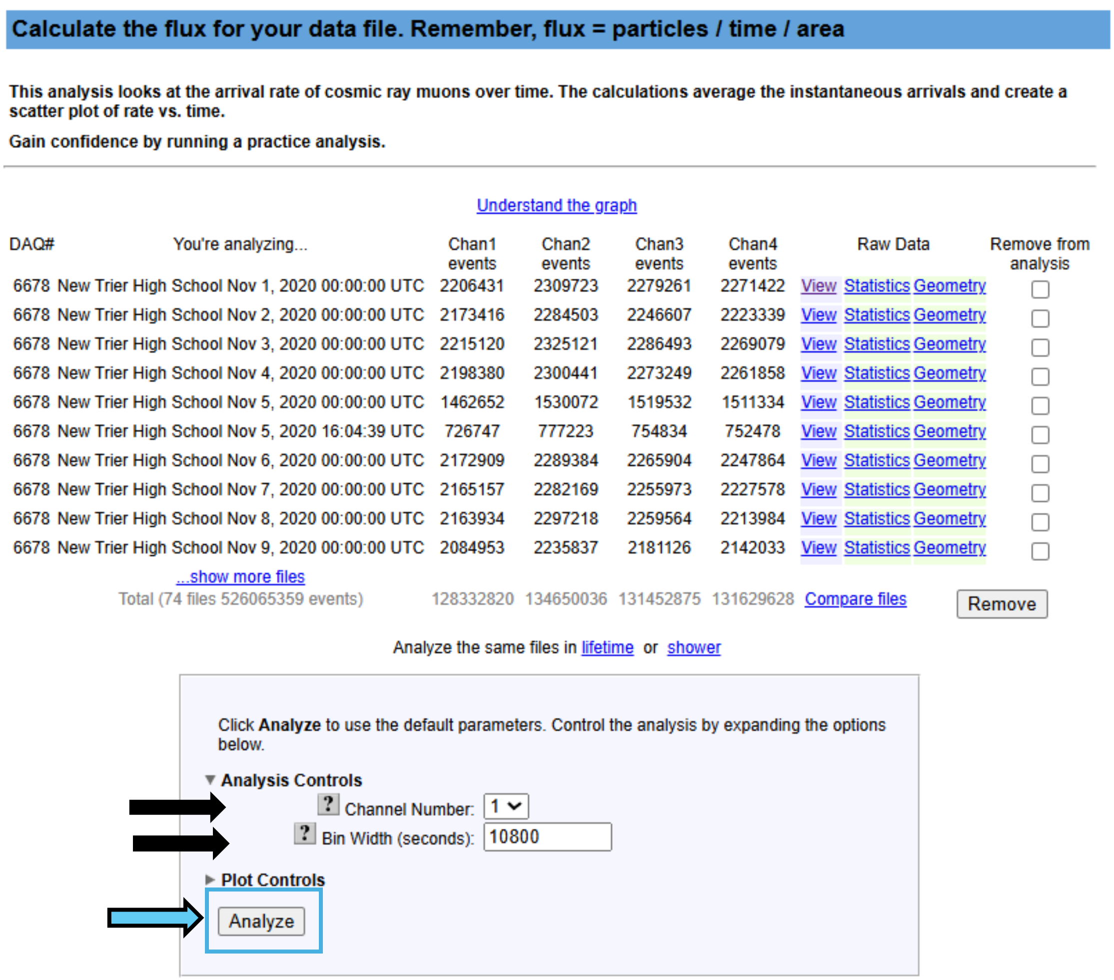
On the right side of the screen, select **Run flux study.**



In this example, we are choosing *Channel Number* ***1*** and changing the default *Bin Width (seconds)* of 600 to **10800** (3-hour bins). This value worked for our data analysis which matched the reporting interval of pressure data from a nearby weather station. Press the **Analyze** button.

The data range for baseline should be done when the sun is quiescent.

This data analysis run may take a while to process. This example took under 15 minutes.



Calculate the flux for your data file. Remember, flux = particles / time / area

This analysis looks at the arrival rate of cosmic ray muons over time. The calculations average the instantaneous arrivals and create a scatter plot of rate vs. time.

Gain confidence by running a practice analysis.

Understand the graph

DAQ#	You're analyzing...	Chan1 events	Chan2 events	Chan3 events	Chan4 events	Raw Data	Remove from analysis
6678	New Trier High School Nov 1, 2020 00:00:00 UTC	2206431	2309723	2279261	2271422	View Statistics Geometry	
6678	New Trier High School Nov 2, 2020 00:00:00 UTC	2173416	2284503	2246607	2223339	View Statistics Geometry	
6678	New Trier High School Nov 3, 2020 00:00:00 UTC	2215120	2325121	2286493	2269079	View Statistics Geometry	
6678	New Trier High School Nov 4, 2020 00:00:00 UTC	2198380	2300441	2273249	2261858	View Statistics Geometry	
6678	New Trier High School Nov 5, 2020 00:00:00 UTC	1462652	1530072	1519532	1511334	View Statistics Geometry	
6678	New Trier High School Nov 5, 2020 16:04:39 UTC	726747	777223	754834	752478	View Statistics Geometry	
6678	New Trier High School Nov 6, 2020 00:00:00 UTC	2172909	2289384	2265904	2247864	View Statistics Geometry	
6678	New Trier High School Nov 7, 2020 00:00:00 UTC	2165157	2282169	2255973	2227578	View Statistics Geometry	
6678	New Trier High School Nov 8, 2020 00:00:00 UTC	2163934	2297218	2259564	2213984	View Statistics Geometry	
6678	New Trier High School Nov 9, 2020 00:00:00 UTC	2084953	2235837	2181126	2142033	View Statistics Geometry	
	...show more files						
	Total (74 files 526065359 events)	128332820	134650036	131452875	131629628	Compare files	Remove

Analyze the same files in lifetime or shower

Click **Analyze** to use the default parameters. Control the analysis by expanding the options below.

Analysis Controls

? Channel Number: 1

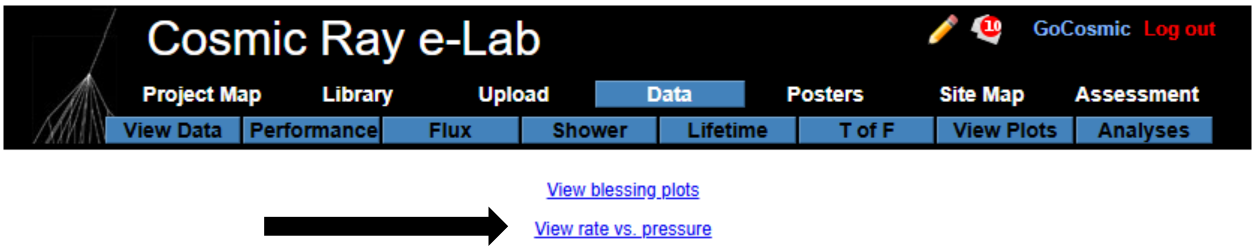
? Bin Width (seconds): 10800

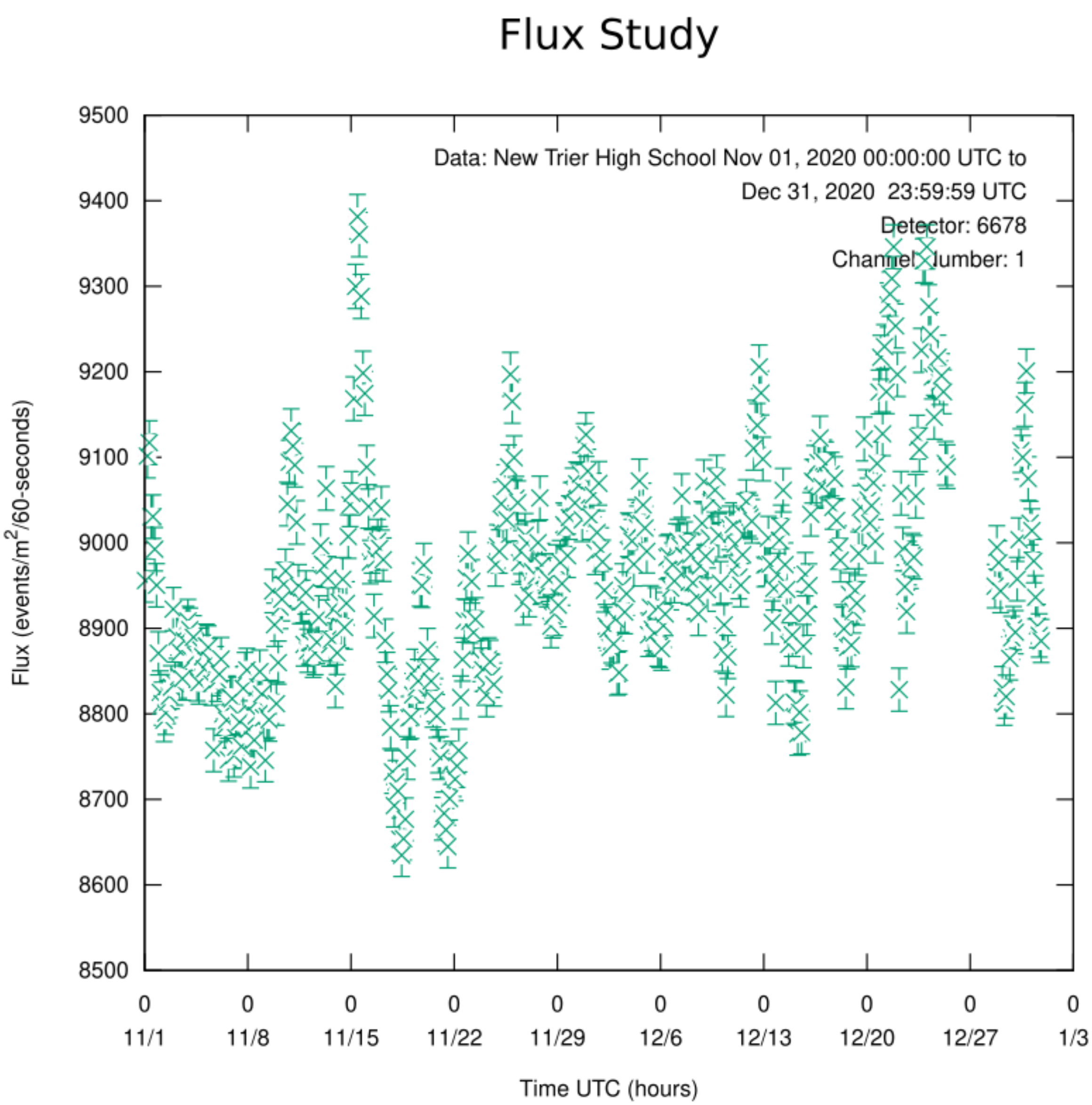
Plot Controls

Analyze

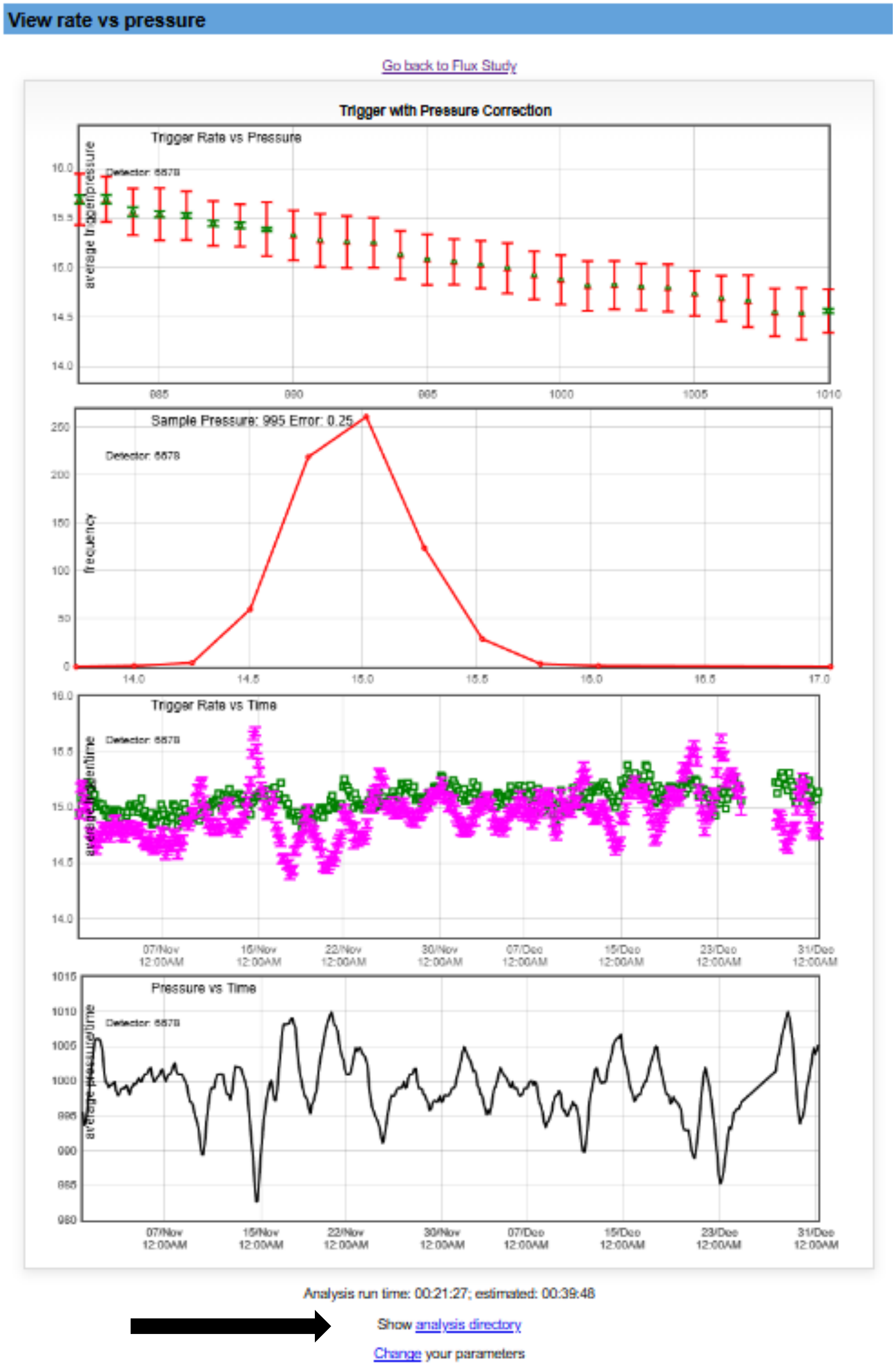
When the analysis is finished, the output is shown below.

Select *View rate vs. pressure*





Select ***analysis directory.***



➢ Open a blank spreadsheet. In the example here, we will use Excel.

We select *flux.out*.

[Note: some lines have been omitted]

Output directory for I2U2.Cosmic::FluxStudy

6678.2020.1105.1.wd
6678.2020.1130.1.wd
6678.2020.1207.1.wd

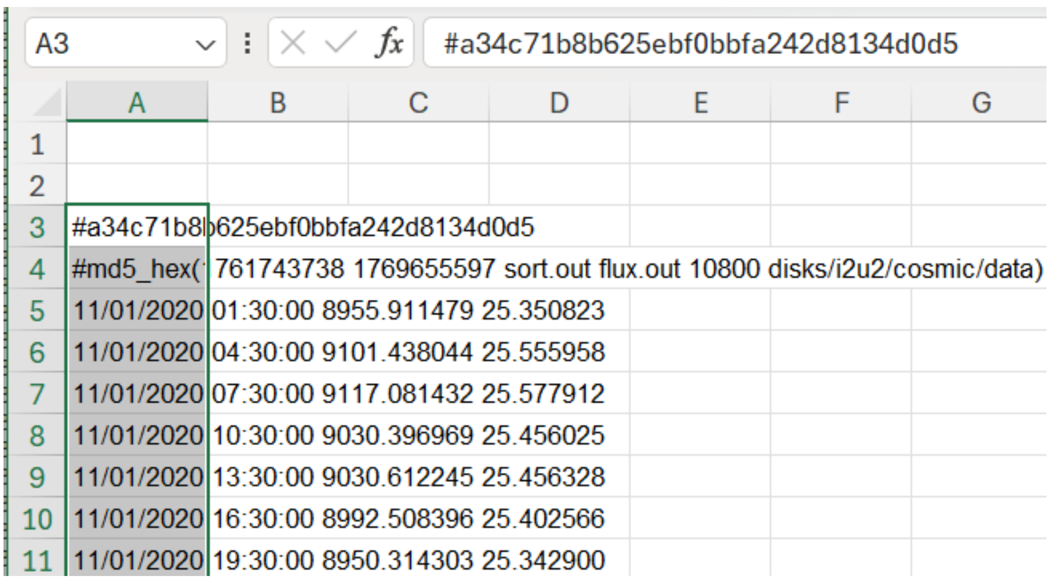
6678.2020.1209.0.wd
6678.2020.1220.0.wd
6678.2020.1211.0.wd
combine.out
singlechannelOut
sort.out
flux.out
plot_param
plot.svg
plot_thm.png
plot.png
dv.dot
RatePressureFlotPlot
RatePressureValues
RateAvgOverPressureValues
RateCorrectionTable

FLUX OUT

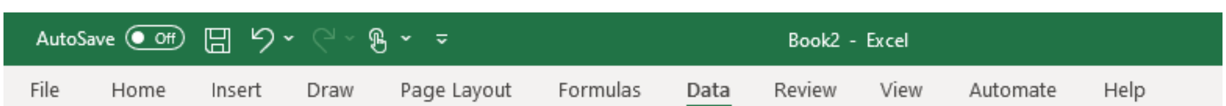
Flux.out (a small portion is shown here). Select all, and copy.

```
#a34c71b8b625ebf0bbfa242d8134d0d5
#md5_hex(1761743738 1769655597 sort.out flux.out 10800 disks/i2u2/cosmic/data)
11/01/2020 01:30:00 8955.911479 25.350823
11/01/2020 04:30:00 9101.438044 25.555958
11/01/2020 07:30:00 9117.081432 25.577912
11/01/2020 10:30:00 9030.396969 25.456025
11/01/2020 13:30:00 9030.612245 25.456328
11/01/2020 16:30:00 8992.508396 25.402566
11/01/2020 19:30:00 8950.314303 25.342900
11/01/2020 22:30:00 8870.805706 25.230084
```

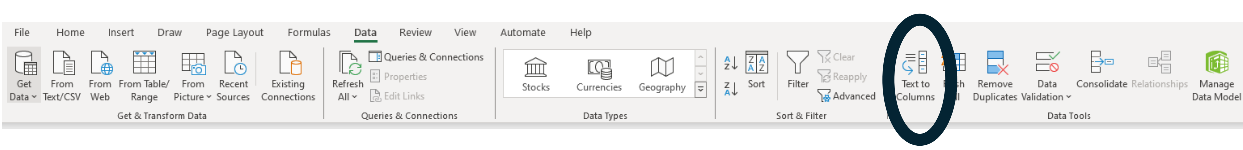
Copy all and paste into a spreadsheet. Pasting into A3 gives room for headers.



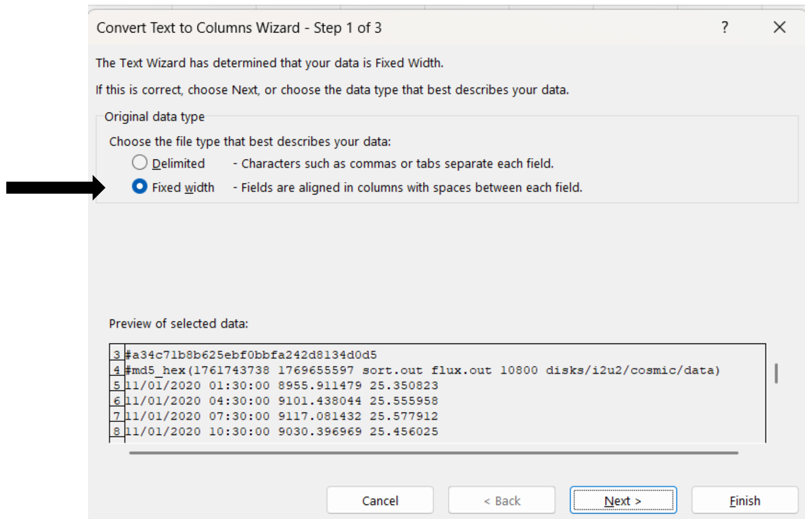
The data need to be divided into columns. From the **menu bar**, select the **Data tab**.



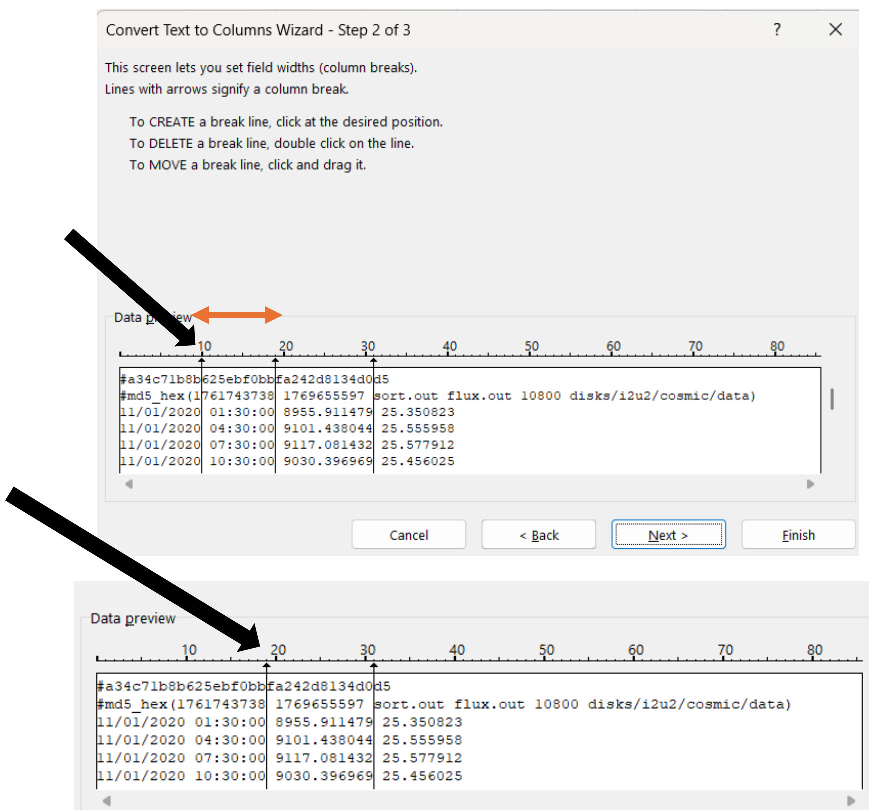
In Data Tools, select *Text to Columns*.



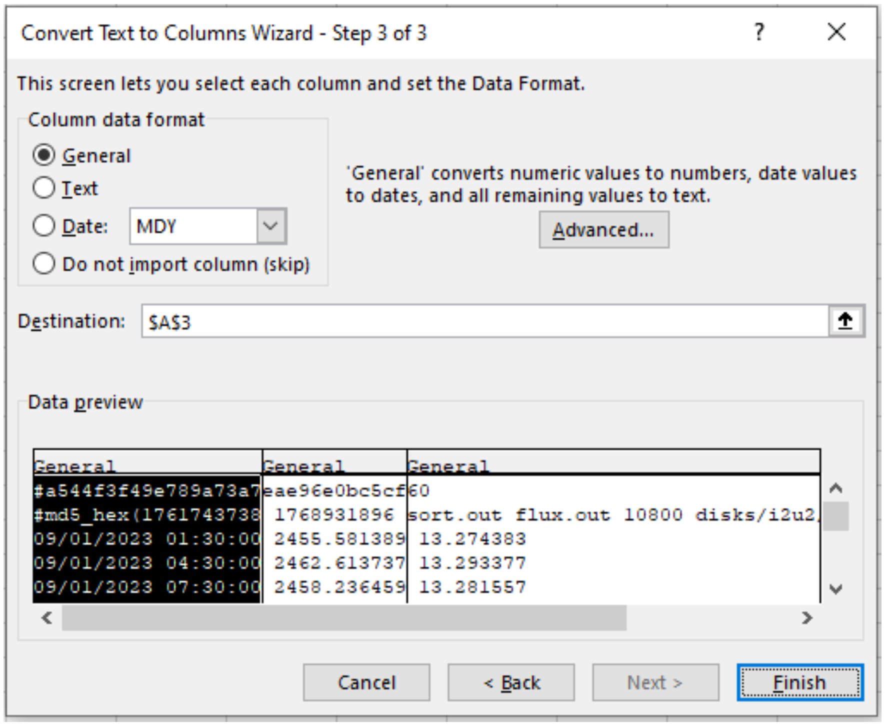
Select *Fixed width*, then *Next>*.



Click and drag this first line to overlap with the second, then *Next>*. This places date and time together.



Check that the first column has both Date and Time, then *Finish*.



This is what the first few rows of the Excel sheet should look like.

	A	B	C	D	E	F	G
1							
2							
3	#a34c71b8b625ebf0bb	fa242d813	d5				
4	#md5_hex(176174373{	1.77E+09	sort.out flux.out 10800 disks/i2u2/cosmic/data)				
5	11/1/2020 1:30	8955.911	25.35082				
6	11/1/2020 4:30	9101.438	25.55596				
7	11/1/2020 7:30	9117.081	25.57791				
8	11/1/2020 10:30	9030.397	25.45603				

Delete Column C, as it is not used. Designate title and headers. Shown are suggestions.

	A	B	C	D	E
1	**Flux 3-Hour Bins For Barometric Correction**				
2					
3	**DATE (mm/dd/yy)** **Time of Center of 3-hour bin (UTC)**	**Flux** **(events/m*m/60sec)**			
4	9/1/2023 1:30	2455.581389			
5	9/1/2023 4:30	2462.613737			
6	9/1/2023 7:30	2458.236459			
7	9/1/2023 10:30	2450.558282			

Pressure Values

► (Note: if your DAQ barometer does not have reliable pressure data, please see the Appendix for instructions on importing non-detector data.)

Return to the *View rate vs pressure* page → [*analysis directory*]() as shown above.

In our example, repeated here, select *RatePressureValues*

Output directory for I2U2.Cosmic::FluxStudy

6678.2020.1105.1.wd
6678.2020.1130.1.wd
6678.2020.1207.1.wd

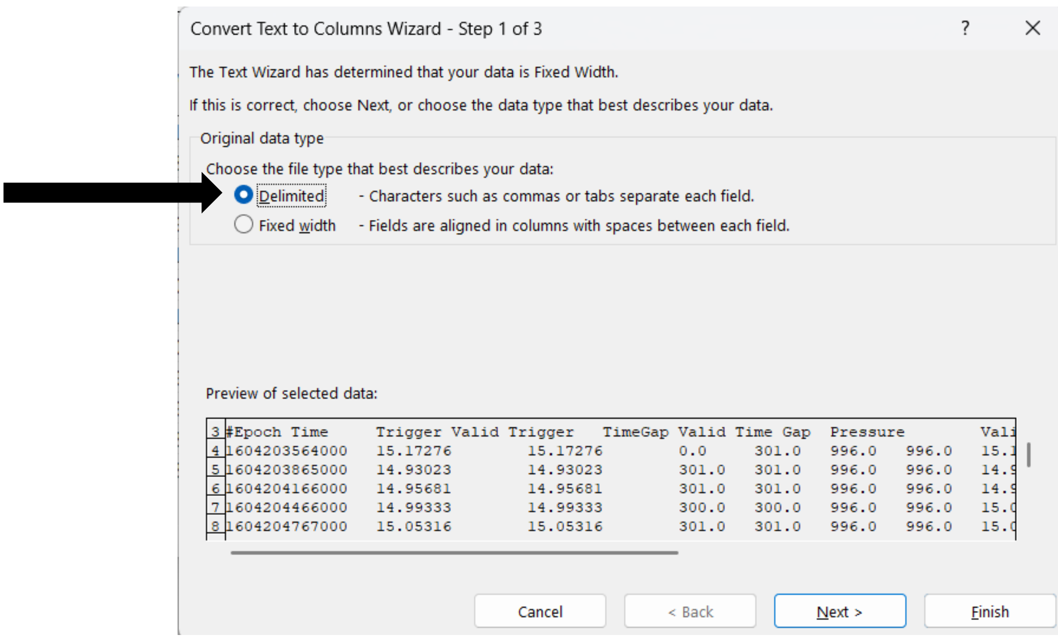
6678.2020.1209.0.wd
6678.2020.1220.0.wd
6678.2020.1211.0.wd
combine.out
singlechannelOut
sort.out
flux.out
plot_param
plot.svg
plot_thm.png
plot.png
dv.dot
RatePressureFlotPlot
RatePressureValues
RateAvgOverPressureValues
RateCorrectionTable

A portion of the table is shown here.

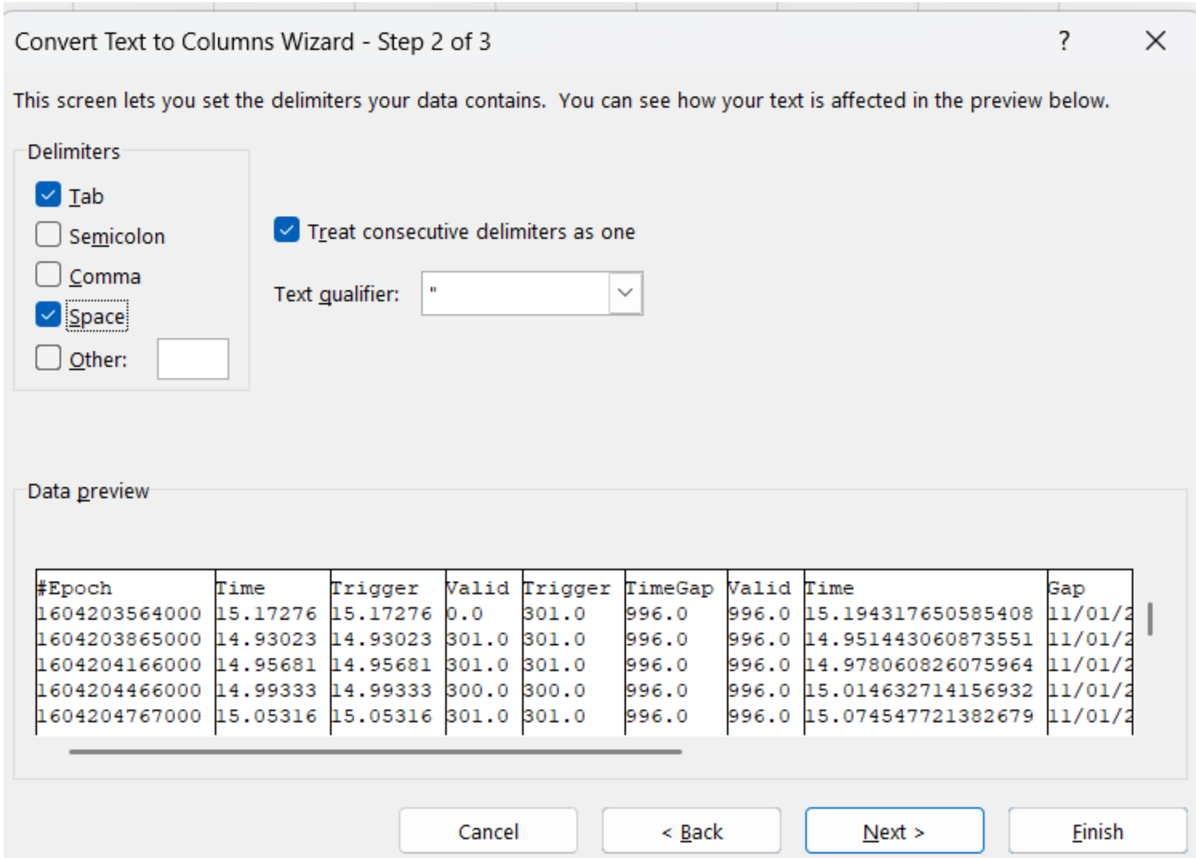
```
#Epoch Time     Trigger Valid Trigger   TimeGap Valid Time Gap  Pressure        Valid Pressure  Corrected Trigger       Date    Seconds
1604203564000   15.17276        15.17276        0.0     301.0   996.0   996.0   15.194317650585408      11/01/2020      364
1604203865000   14.93023        14.93023        301.0   301.0   996.0   996.0   14.951443060873551      11/01/2020      665
1604204166000   14.95681        14.95681        301.0   301.0   996.0   996.0   14.978060826075964      11/01/2020      966
1604204466000   14.99333        14.99333        300.0   300.0   996.0   996.0   15.014632714156932      11/01/2020      1266
1604204767000   15.05316        15.05316        301.0   301.0   996.0   996.0   15.074547721382679      11/01/2020      1567
1604205068000   14.70764        14.70764        301.0   301.0   996.0   996.0   14.728536802167566      11/01/2020      1868
1604205369000   14.92691        14.92691        301.0   301.0   996.0   996.0   14.948118343775281      11/01/2020      2169
```

Select all, copy, and paste into the Excel file that was created. Paste into Cell F3.

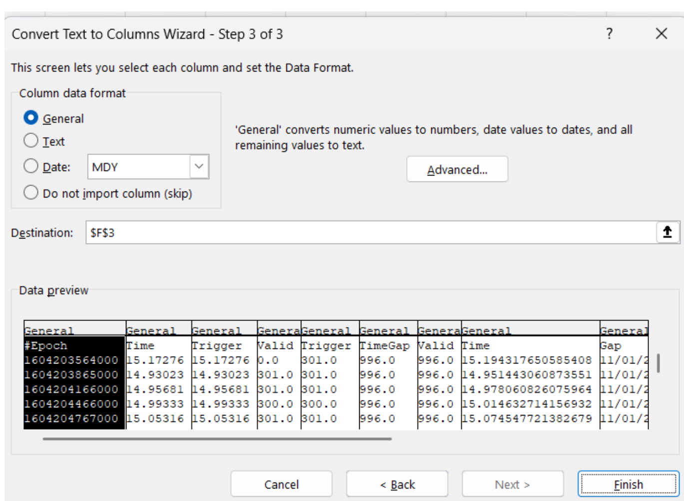
In Data Tools, select *Text to Columns.* Select Delimited, then *Next*.



Place a checkmark in the Space box, then *Next*



Then press *Finish*.



Adjust cell width and column formatting as desired.

F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U
#Epoch	Time	Trigger	Valid	Trigger	TimeGap	Valid	Time	Gap	Pressure	Valid	Pressure	Corrected	Trigger	Date	Seconds
1693541251000	12.18	12.18	0	301	999	999	12.30	9/1/2023	451						
1693541552000	12.35	12.35	301	301	999	999	12.47	9/1/2023	752						
1693541853000	12.22	12.22	301	301	999	999	12.34	9/1/2023	1053						

Designate title and headers. Shown are suggestions.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	**Flux 3-Hour Bins For Barometric Correction**					**Barometric Pressure Data Every 300 Seconds From Start of Experiment Run**									
2															
3	**DATE (mm/dd/yy) Time of Center of 3-hour bin (UTC)**	**Flux (events/m*m/60sec)**				**#Epoch Time**	**Trigger**	**Valid Trigger**	**Time Gap**	**Valid Time Gap**	**Pressure**	**Valid Pressure**	**Corrected Trigger**	**Date mm/dd/yyyy**	**Seconds after Midnight**
4	11/1/2020 1:30	8956				1604203564000	15.17	15.17	0	301	996	996	15.19	11/1/2020	364
5	11/1/2020 4:30	9101				1604203865000	14.93	14.93	301	301	996	996	14.95	11/1/2020	665
6	11/1/2020 7:30	9117				1604204166000	14.96	14.96	301	301	996	996	14.98	11/1/2020	966
7	11/1/2020 10:30	9030				1604204466000	14.99	14.99	300	300	996	996	15.01	11/1/2020	1266
8	11/1/2020 13:30	9031				1604204767000	15.05	15.05	301	301	996	996	15.07	11/1/2020	1567

Note: Do not merge the cells for the titles *Flux 3-Hour Bins For Barometric Correction* and *Barometric Pressure Data Every 300 Seconds From Start of Experiment Run*. Grouped cells can be difficult to work with. Formatting of titles can be done later. Add the DAQ number as shown in the next screen shot.

- The flux data (Column B) are in three-hour bins, starting at midnight, and is reported at the time of the middle of the bin.
- The barometric pressure (Column K) is in 5-minute increments from whenever the data run was started, not midnight. The offset after midnight, in seconds, is in Column O.
- Three hours is 10,800 seconds.

We write the times for the starts and ends of each bin based on the time in Column A. Insert 2 columns between A and B and title them *Bin Start Time* and *Bin End Time*, respectively. For the start times you need to subtract 1.5 hours, and for the end times you need to add 1.5 hours†. In Cell B4 enter **=$A4-(1.5/24)** and Cell C4 enter **=$A4+(1.5/24).** Fill in the rest of the times‡. Make sure these times match up with what you expect.

†If you are unfamiliar with Excel, dates and times are just numbers (in days) that Excel formats to look like a date. If we want to add 1.5 hours to a time, we need to divide the value by 24 since there are 24 hours in a day.

‡If you are unfamiliar with Excel, you do not need to scroll all the way down to select all the cells. Use ctrl+shift+down or ctrl+down to select and navigate cells.

Adjust cell width and column formatting as desired.

	A	B	C	D
1	**Flux 3-Hour Bins For Barometric Correction**			
2	**DAQ 6678**	$A4-1.5/24	$A4+1.5/24	
3	**DATE (mm/dd/yy)** **Time of Center of 3-hour bin (UTC)**	**Bin Start Time**	**Bin End Time**	**Flux (events/m*m/60 sec)**
4	11/1/2020 1:30	11/1/2020 0:00	11/1/2020 3:00	8956

We need to find the times associated with the pressure readings. The column labeled *Seconds after Midnight* (Column **Q)**, and the column labeled *Date* (Column P) are needed for the next step. In Column G, we are going to write the time*. In Cell G4, write: =$P4+($Q4/24/60/60). Format the column so it shows the date *and time*. Fill down.

**For the same reason as before, we need to divide the number by 24/60/60 to convert it from days to hours to minutes to seconds.*

G	H	I	J	K	L	M	N	O	P	Q
	Barometric Pressure Data Every 300 Seconds From Start of Experiment Run									
$P4+($Q4/24/60/60)										
UTC time	**#Epoch Time**	**Trigger**	**Valid Trigger**	**Time Gap**	**Valid Time Gap**	**Pressure**	**Valid Pressure**	**Corrected Trigger**	**Date mm/dd/yyyy**	**Seconds after Midnight**
11/1/20 0:06	1604203564000	15.17	15.17	0	301	996	996	15.19	11/1/2020	364
11/1/20 0:11	1604203865000	14.93	14.93	301	301	996	996	14.95	11/1/2020	665

In Cell E3, label the cell *Average Pressure 3-hour bins.*

	A	B	C	D	E
1	**Flux 3-Hour Bins For Barometric Correction**				
2	**DAQ 6678**	$A4-1.5/24	$A4+1.5/24		
3	**DATE (mm/dd/yy)** **Time of Center of 3-hour bin (UTC)**	**Bin Start Time**	**Bin End Time**	**Flux (events/m*m/60 sec)**	**Average Pressure 3-hour bins**

Average the pressure readings whose times fall between the start and end times of our bin. In Column E. In Cell E4, write:

=IFERROR(AVERAGEIFS($M:$M,$G:$G,">="&$B4,$G:$G,"<"&C4),"")

Fill down.

Explanation of the equation.

=IFERROR(AVERAGEIFS($M:$M,$G:$G,">="&$B4,$G:$G,"<"&C4),"")

IFERROR returns a value if the rest of the equation returns an error. Otherwise, it just returns the value. In this instance, if the center section is an error, it will return "" which is nothing. If it is not an error, the cell will be:

AVERAGEIFS($M:$M,$G:$G,">="&$B4,$G:$G,"<"&$C4).

AVERAGEIFS returns the average of all cells that meet multiple criteria. In our case, this function averages the cells in Column M (*Pressure*), whose corresponding Cell G value (*UTC Time*) is both greater than or equal to B4 (*Bin Start Time*), and also less than C4 (*Bin End Time*).

It does not matter whether the start or end times are inclusive

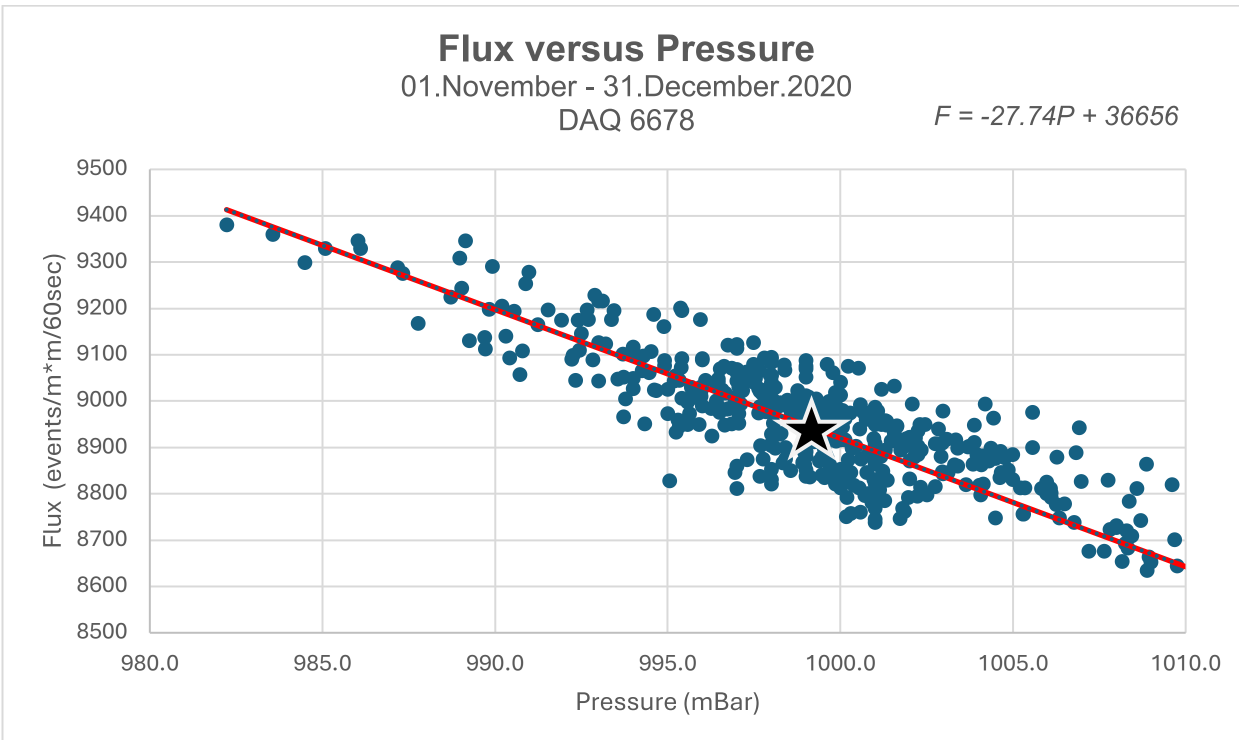
SUM | =IFERROR(AVERAGEIFS($M:$M,$G:$G,">="&$B4,$G:$G,"<"&$C4),"")

	A	B	C	D	E	F
1	**Flux 3-Hour Bins For Barometric Correction**					
2	**DAQ 6678**	$A4-1.5/24	$A4+1.5/24		IFERROR(AVERAGEIFS($M:$M,$G:$G,">="&$B4,$G:$G,"<"&$C4),"")	
3	**DATE (mm/dd/yy)** **Time of Center of 3-hour bin (UTC)**	**Bin Start Time**	**Bin End Time**	**Flux (events/m*m/60sec)**	**Average Pressure 3-hour bins**	
4	11/1/2020 1:30	11/1/2020 0:00	11/1/2020 3:00	8956	=IFERROR(AVERAGEIF	
5	11/1/2020 4:30	11/1/2020 3:00	11/1/2020 6:00	9101	993.7	
6	11/1/2020 7:30	11/1/2020 6:00	11/1/2020 9:00	9117	994.0	
7	11/1/2020 10:30	11/1/2020 9:00	11/1/2020 12:00	9030	995.9	
8	11/1/2020 13:30	11/1/2020 12:00	11/1/2020 15:00	9031	998.1	

	G	H	I	J	K	L	M	N	O	P	Q
1	**Barometric Pressure Data Every 300 Seconds From Start of Experiment Run**										
2	$P4+($Q4/24/60/60)										
3	**UTC time**	**#Epoch Time**	**Trigger**	**Valid Trigger**	**Time Gap**	**Valid Time Gap**	**Pressure**	**Valid Pressure**	**Corrected Trigger**	**Date mm/dd/yyyy**	**Seconds after Midnight**
4	11/1/20 0:06	1604203564000	15.17	15.17	0	301	996	996	15.19	11/1/2020	364
5	11/1/20 0:11	1604203865000	14.93	14.93	301	301	996	996	14.95	11/1/2020	665
6	11/1/20 0:16	1604204166000	14.96	14.96	301	301	996	996	14.98	11/1/2020	966
7	11/1/20 0:21	1604204466000	14.99	14.99	300	300	996	996	15.01	11/1/2020	1266
8	11/1/20 0:26	1604204767000	15.05	15.05	301	301	996	996	15.07	11/1/2020	1567

We now graph Flux (Column D) versus Pressure (Column E). To make things less cluttered, we have created a new tab, Flux vs Pressure.

Shown here is a linearization of a scatter plot.



Slope = -27.736

Using the MEDIAN formula, plot the median of Pressure and Flux

Average Pressure 3-hour bins	Flux (events/m*m/60sec)	
999.1	8952.9	Median

★ Median on graph

Correction of Flux due to Barometric Pressure

We need to solve the equation below at every point.

$$F_{corrected} = F_{Original}\left\{1+\left[\left(\frac{|m|}{F_{median}}\right)*(P-P_{median})\right]\right\}$$

$F_{corrected}$= Corrected Flux

$F_{original}$ = Uncorrected Flux

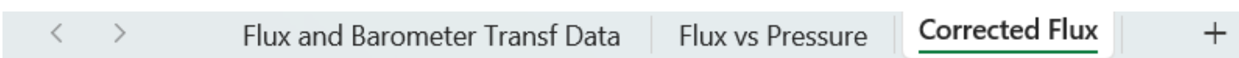
m = slope of flux vs Pressure using a control month (K_p index < 4) [absolute value]

F_{median}= median flux where the $P = P_{median}$ on F vs P graph

P = pressure at $F_{original}$

P_{median} = median pressure value on Control F vs P Graph

For corrected flux, a new sheet, *Corrected Flux*, was created to keep spreadsheets orderly.



Constants and formula are on the left, and the data:

Column E (='Flux and Barometer Transf Data'!D3:D653)

Column F (='Flux and Barometer Transf Data'!Q3:Q652)

are copied from other sheets.

The corrected flux is calculated here

=E4*(1+((ABS($C$4)/$C$5)*(F4-$C$6)))

SUM | =E6*(1+((ABS($C$4)/$C$5)*(F6-$C$6)))

	A	B	C	D	E	F	G	H	I
1	**Calculations Corrected Flux DAQ 6678**								
2					Original			Corrected	
3					Flux (events/m*m/60sec)	Average Pressure 3-hour bins		Flux (events/m*m/60sec)	DATE (mm/dd/yy) Time of Center of 3-hour bin (UTC)
4	Slope of Flux versus Pressure	m	-27.74		8956	995.6		8860	11/1/2020 1:30
5	Median Flux	Fmedian	8953		9101	993.7		8950	11/1/2020 4:30
6	Median Pressure	Pmedian	999.1		9117	994.0		=E6*(1+((	11/1/2020 7:30
7					9030	995.9		8943	11/1/2020 10:30
8					9031	998.1		9004	11/1/2020 13:30
9	Corrected Flux Formula				8993	1000.6		9035	11/1/2020 16:30
10	E4*(1+((ABS($C$4)/$C$5)*(F4-$C$6)))				8950	1002.3		9040	11/1/2020 19:30
11					8871	1004.3		9015	11/1/2020 22:30
12					8825	1006.0		9014	11/2/2020 1:30

The Date and Time (UTC) need to be copied to Column I of *Corrected Flux* from Column A of *Flux and Barometer Transf Data*.

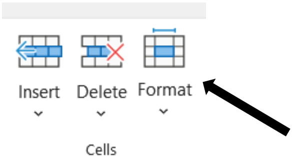
= 'Flux and Barometer Transf Data'!A3:A653

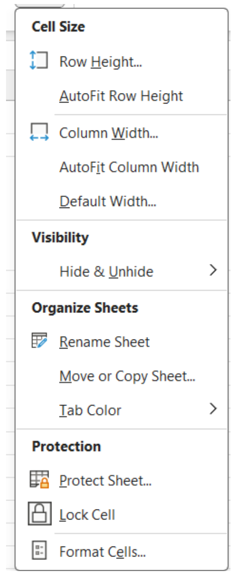
If the dates are not being displayed as date/time, change the formatting of that cell (column) to be mm/dd/yyyy h:mm. (Another item to check is to ensure the correct column is being copied.)

	A
1	**Flux 3-Hou**
2	
3	DATE (mm/dd/yy) Time of Center of 3-hour bin (UTC)
4	9/1/2023 1:30
5	9/1/2023 4:30
6	9/1/2023 7:30
7	9/1/2023 10:30
8	9/1/2023 13:30

I
DATE (mm/dd/yy) Time of Center of 3-hour bin (UTC)
45170
45170
45170
45170
45171

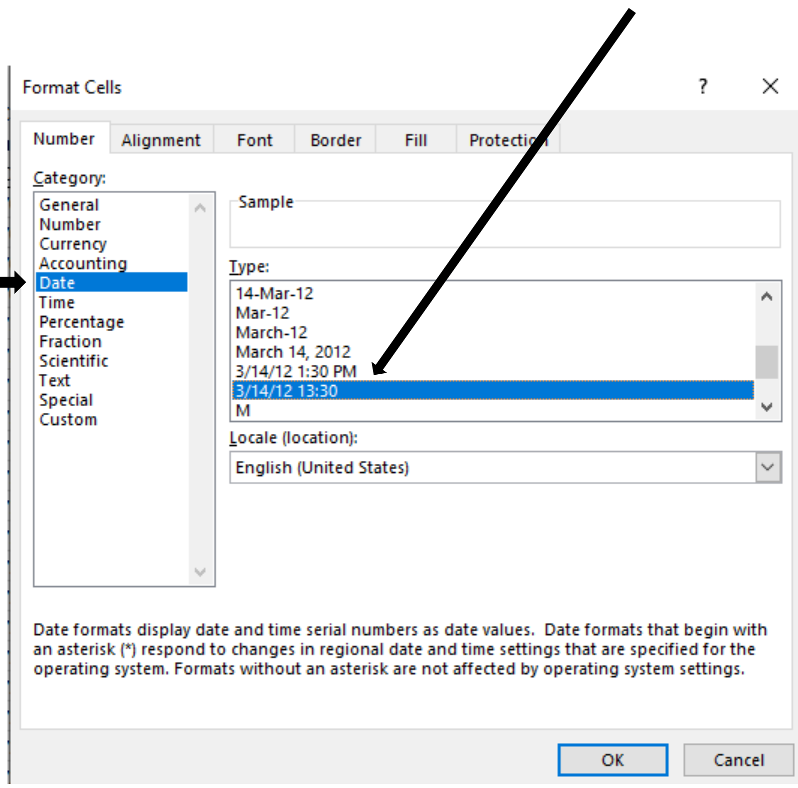
You may need to format these as a date. Highlight Column I and select Format Cells from the Home tab.





Select Format Cells

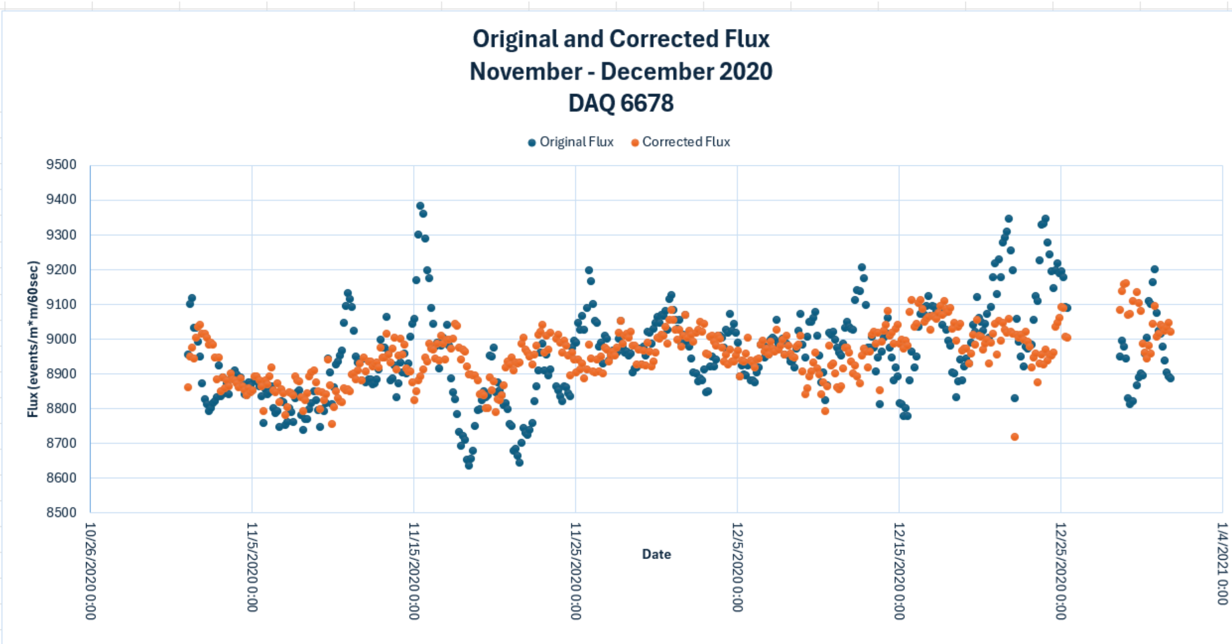
Select *Date* from left side. In the right window, scroll and select 3/14/12 13:30, then press OK.



The date and time formatted correctly.

H	I
Corrected	
Flux (events/m*m/60sec)	DATE (mm/dd/yy) Time of Center of 3-hour bin (UTC)
8860	11/1/2020 1:30
8950	11/1/2020 4:30
8974	11/1/2020 7:30
8943	11/1/2020 10:30
9004	11/1/2020 13:30

The corrected flux versus time graph is plotted. Error bars can be added.



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